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Atal Narayan Sahu

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Experience

UCLA | *Graduate Research Assistant, Quantum Light Matter Cooperative* — Mar. 2026 – Present

- Developing a domain-specialized Qwen2.5-Coder-14B for PennyLane code generation using LoRA-based continued pretraining and a post-training pipeline of DPO and GRPO with rewards verified by quantum circuit simulators.
- Developed a PyTorch diffusion model with a ResNet–BiLSTM–attention encoder to reconstruct pulse amplitude and phase from sparse spectrograms, achieving 13x lower amplitude error than SOTA; presented at Optica APC 2026.

PrashantAdvait Foundation | *Senior Software Engineer (Machine Learning)* — Apr. 2024 – Apr. 2025

- Increased time spent per active user by 62% on Gita Feed (20K+ DAU) by owning the candidate-selection pipeline and an XGBoost learning-to-rank model using content richness, user interaction history, and post metadata.
- Led development of AskAP, a bilingual English–Hindi RAG assistant combining embedding search, lexical matching, and reranking to generate source-attributed answers from 20K documents and serve 8K queries daily.
- Architected a Couchbase-to-ClickHouse ELT pipeline via materialized views, enabling sub-100 ms dashboard latency and sub-1 minute refresh SLAs for outreach operations handling 100K+ calls/day.

Regology, Inc. (acquired by Bloomberg) | *Senior Data Scientist* — Sep. 2021 – Mar. 2024

- Led a 4-person team to build and launch Reggi, a legal RAG assistant spanning ingestion, query understanding, hybrid retrieval, LangChain-based generation, multi-turn context management, and monitoring.
- Architected a two-level hierarchical search system in Elasticsearch, combining BM25 lexical retrieval with dense semantic retrieval for 1M+ legal documents across corpora, improving Recall@10 by 5.7x (13% → 74%).
- Implemented domain adaptation for legal bi-encoders in SentenceTransformers through synthetic query generation, hard-negative mining, and pseudo-labeling, improving Recall@10 from 25% to 40% on held-out queries.
- Developed a keyphrase extraction algorithm for hierarchical legal documents, replacing standard TF-IDF with a computationally efficient approach based on corpus statistics and improving downstream Recall@10 by 1.8x.

KAUST | *Graduate Research Assistant, SANDS Lab* — Jan. 2019 – Aug. 2021

- Implemented a distributed optimizer in PyTorch that preserved language-model perplexity on WikiText-2 while reducing gradient communication by 99.8% across 8 GPUs (NeurIPS 2021 Spotlight, Top 3%). [Paper] [Code]
- Developed OmniReduce, a streaming sparse AllReduce system, and designed block-based gradient sparsification methods that accelerated distributed BERT Transformer fine-tuning on SQuAD by 1.7x (ACM SIGCOMM 2021).

INRIA | *Summer Research Intern, STARS Team* — Jun. 2020 – Aug. 2020

- Implemented ordering losses in TensorFlow/Keras for a Temporal CNN, achieving 33x faster convergence (133 → 4 min) with 17.8% lower accuracy than ISBA on weakly supervised video action segmentation. [Report] [Code]

Technical Skills

Languages: Python, SQL, Go, C, Julia
ML, NLP & Search: PyTorch, JAX, TensorFlow, Hugging Face Transformers, TRL, SentenceTransformers, LoRA, DPO, GRPO, RLHF, FSDP, scikit-learn, XGBoost, NumPy, Elasticsearch, LangChain, LangGraph, LlamaIndex, spaCy
Data, Systems & MLOps: AWS, Docker, Ray, Spark, ClickHouse, Couchbase, MongoDB, dbt, W&B, CI/CD

Publications

Publications at NeurIPS, AAAI, SIGCOMM, and EuroSys; see the full list on Google Scholar.

Education

UCLA | *Master of Quantum Science & Technology (Focus: AI for Science)* — Sep. 2025 – Sep. 2026
KAUST | *M.S. in Computer Science (Focus: Distributed ML)* — Jan. 2019 – Dec. 2020
IIT Kanpur | *B.S. in Mathematics & Scientific Computing; Minors in Algorithms & ML* — Jul. 2014 – Jun. 2018